

Highlights

TURRET OS: Multi-Layered Graph and Rule-AST Architecture Rescuing Insider Threat Detection Against Out-of-Distribution Adversaries

- First 4-layer architecture for OOD insider threat detection.
- Rescues OOD credential phishing from 26.25% drop down to 4.12%.
- Achieves 100% attribute coverage for ISO/IEC 27043 readiness.
- Emits Ed25519-signed W3C-PROV evidence packs with dual hashing.
- Reports ROC-AUC 0.9997 on D3 TSEC & 0.9928 on D5 OOD hold-out.